Optimizing Agricultural Resource Allocation through Reinforcement Learning for Yield Improvement: A Cost-Driven Approach to Crop Efficiency Enhancement

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Abstract

This synopsis proposes a Pakistan-adapted reinforcement learning (RL) framework to optimize agricultural resource allocation (water, nitrogen) under monsoon variability. We build **CyclesGym-PK** on top of the Cycles agroecosystem model and OpenAI Gym-style interfaces to learn weekly irrigation/fertilization policies that maximize *profit* while constraining *environmental losses*. The study addresses three gaps:

- Limited transferability of existing RL environments (largely temperate, single-season, greenhouse-focused)
- Lack of localized data (weather, soils, costs) in prior work
- Absence of long-horizon, multi-objective optimization for Pakistan's open-field, smallholder context.

We specify the environment, reward design, evaluation protocol (dry/normal/wet years; soil and price shocks), and KPIs (yield, mm irrigation, PKR profit, N-use efficiency, leaching). Expected outcomes include reduced input waste, improved profitability, and a reproducible pipeline for region-specific policy learning.

Chapter 1

Introduction

1.1 Background and Motivation

Agriculture drives employment and GDP in Pakistan, yet yields and water-use efficiency trail global frontiers. Static heuristics (e.g., fixed irrigation intervals) underperform when monsoon timing shifts. Simulation-first RL offers a cost- and time-efficient way to explore thousands of management strategies virtually, prior to field pilots, reducing risk and accelerating learning.

1.2 Problem with Field Hit-and-Trial

Field experimentation is slow, costly, and risky. Each trial consumes a season and real inputs; failures cause real losses. By contrast, virtual experiments can compress years into hours, but demand *credible simulators* and localized data.

1.3 Advantages of Simulation-Based RL

- **Cost/Time:** Thousands of episodes without physical inputs.
- **Scalability:** Diverse soils/crops/weather can be sampled.
- **Safety:** No real-world crop loss during policy search.

A direct comparison of these two approaches is provided in Table 1.1. Recent data, shown in Figure 1.1 and Figure 1.2, highlights these yield gaps. Key constraints are summarized in Figure 1.3.

Table 1.1: Simulation RL vs. Field Hit-and-Trial

Aspect	**Field Hit-and-Trial**	**Simulation-Based RL**
Cost	High (inputs, labor, land)	Low (compute only)
Time	One season per trial	Many seasons/hour
Risk	Crop loss possible	No real loss
Scale	Logistics limited	Virtually unlimited
Data	Field observations	Requires accurate models/data

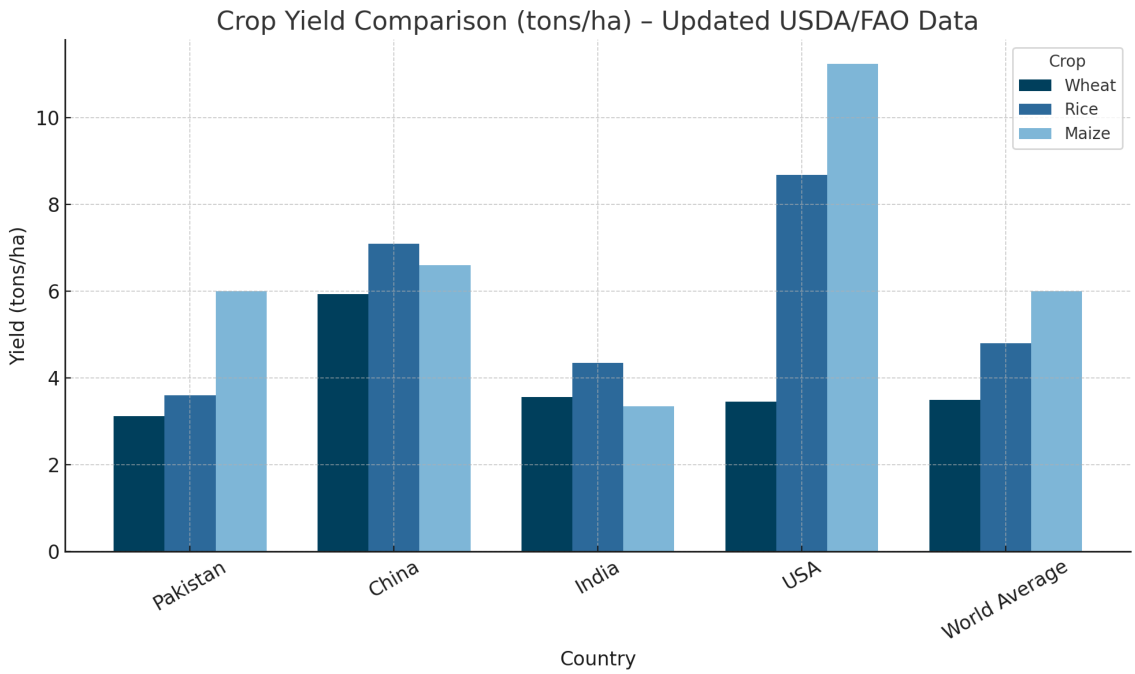

Figure 1.1: Crop Yield Comparison (tons/ha) – Updated USDA/FAO Data.

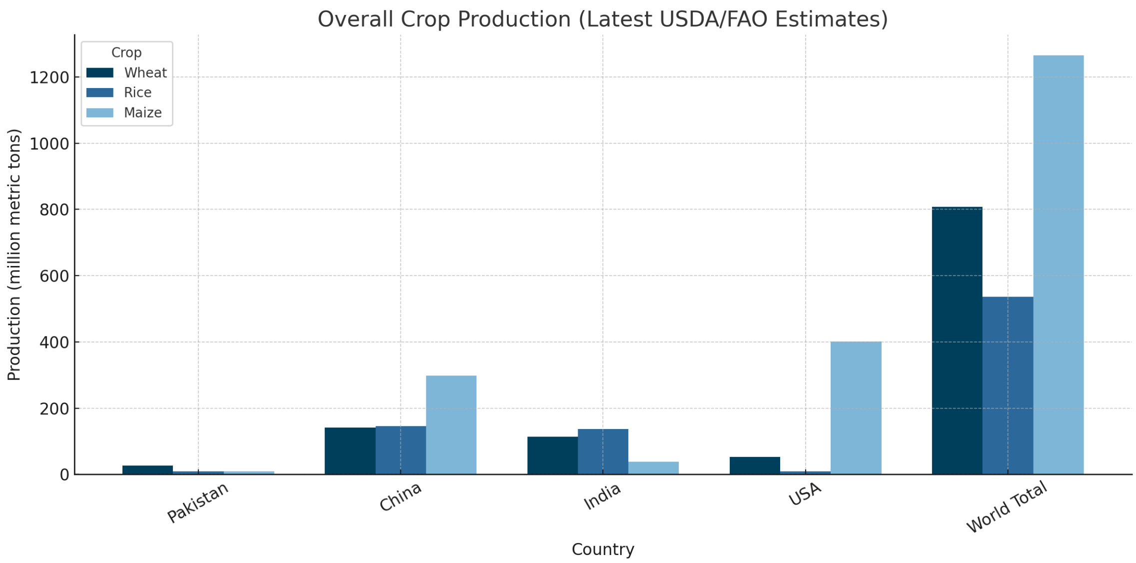

Figure 1.2: Overall Crop Production (Latest USDA/FAO Estimates).

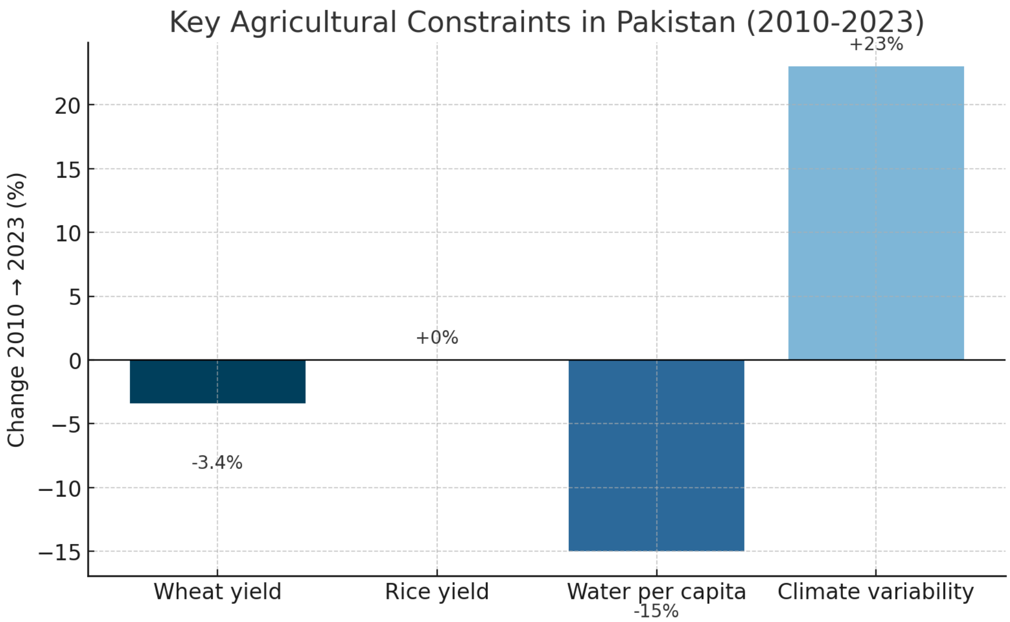

Figure 1.3: Key Agricultural Constraints in Pakistan (2010–2023).

Chapter 2

Literature Review

2.1 Scope and Organization

This review covers three strands:

- **Machine Learning (ML)** for agricultural prediction and decision support
- **Deep Learning (DL)** for perception and controlled-environment control
- **Reinforcement Learning (RL)** for sequential, long-horizon crop management.

2.2 Machine Learning for Agriculture: strengths and limits

Hybrid crop-model + ML prediction (Frontiers 2023). Dhillon et al. integrate a light-use-efficiency (LUE) crop model with Random Forests to predict winter wheat and oilseed rape yields; the hybrid improves R^2 and RMSE versus pure ML across Bavaria, Germany (Frontiers in Remote Sensing 2023) [1]. The merit is fusing physiological priors with data; the limitation is regional calibration burden and the fact that the output is *prediction*, not a prescriptive policy.

Clustered ensembles for heterogeneous smallholders (AAAI 2022). Tabar et al. propose clustered ensembles to handle diverse African farm time-series; errors drop versus LSTM/VRNN baselines (AAAI 2022) [2]. However, labels/clusters need curation and the method forecasts stress/yield rather than *deciding* actions.

Sensor minimization via GBM (Frontiers in Plant Science 2022). Uyeh et al. use Gradient Boosting and psychrometric features to choose minimal greenhouse sensors with low RMSE (Frontiers in Plant Science 2022) [3]. Applicable to cost-sensitive setups, but limited to temperature/humidity in two greenhouses; generality is unclear.

Explainable crop recommendation (Scientific Reports 2025). Shastri et al. compare 10 ML algorithms, with Gradient Boosting strongest; LIME explains feature influence (Sci Rep 2025) [4]. The merit is transparency; the limitation is single-season, static recommendations on India-specific data.

2.3 Deep Learning for perception/control: value and boundaries

Aerial weed manipulation (WACV 2023). Steininger et al. release the CropAndWeed dataset (8k images; 112k instances) and CNN baselines, boosting precision weeding (WACV 2023) [5]. Useful for automated actuation, but relies on drone views; ground-level/weather robustness remains open.

Autonomous greenhouse control with deep RL (AAAI 2022). Cao et al. (iGrow) build a neural "digital twin" and deep-RL controller achieving +10% tomato yield and +92% profit over experts in controlled settings (AAAI 2022) [6]. This evidences data-driven control benefits; yet calibration is environment-specific and generalization to open fields is unproven.

Additional DL exemplars. Edge-deployable leaf disease detection (Computers & Electronics in Agriculture 2022) [7], augmentation-robust aerial segmentation (CVPRW 2022) [8], and vision–language few-shot disease detection (AAAI 2025) [9] all expand sensing capability; however, most target perception, not multi-season resource optimization in open fields.

2.4 Reinforcement Learning for sequential farm management

2.4.1 Single-season RL environments

CropGym (Env Data Sci 2023). Built atop PCSE/LINTUL-3 for winter-wheat nitrogen scheduling with 18D state and a β trade-off between yield and fertilizer cost (EDS 2023) [10]. *Demerits for our objective:* primarily single-season; limited explicit soil carryover; narrow action scope.

Gym-DSSAT (Inria RR 9460, 2022). Gym interface around DSSAT via Fortran–Python coupling (PDI), enabling daily maize decisions with stochastic weather (Inria 2022) [11]. *Demerits:* heavy setup; maize-centric demos; limited multi-year rotation modeling.

2.4.2 Multi-year RL with environmental constraints

CyclesGym (NeurIPS 2022). Multi-year, multi-crop environment based on the Cycles model with nitrogen fixation, rotation dynamics, and modular state/reward/weather generators (NeurIPS D&B 2022) [12]. It supports complex actions (crop type, planting date, fertilizer/irrigation decisions) and explicit environmental constraints (leaching/emissions), aligning with our Pakistan goals.

2.4.3 Why long-term planning matters

Decisions are interdependent across years: fertilizer timing affects next-season soil N; rotations alter pests and water balance. A long-horizon RL objective $\max_\pi \mathbb{E}_\pi\left[\sum_{t=0}^{T} \gamma^t r_t\right]$

with constraints on leaching and budgets encodes realistic trade-offs between profit and sustainability (NeurIPS 2022) [12].

The agent-environment loop (Figure 2.2), a sample decision tree (Figure 2.1), and the Gym-DSSAT architecture (Figure 2.3) illustrate these concepts.

2.5 Synthesis: implications for a Pakistan-adapted framework

ML/DL improve perception and prediction but rarely *decide* weekly actions across seasons. Single-season RL (CropGym, Gym-DSSAT) proves feasibility, yet lacks multi-year rotation/soil carryover central to Indus-plain rice–wheat systems. CyclesGym's rotation and constraint modeling therefore provides the most faithful substrate for **CyclesGym-PK**.

2.6 Comprehensive Comparison Table

Abbreviations: EDS = Environmental Data Science; CEA = Computers & Electronics in Agriculture.

A comprehensive summary of the key literature is presented in Table 2.1.

Table 2.1: Key Literature Across ML, DL, and RL

Title	Venue	Year	Domain	Approach	Strength	Weakness	Limitations
Integrating Random Forest and Crop Modeling Improves Yield of Winter Wheat and Oilseed Rape [1]	Frontiers (Remote Sensing)	2023	ML	LUE crop model + RF	Hybrid raises R^2, lowers RMSE	Region-specific calibration	Predictive (not prescriptive); limited transferability [a]
CLIMATES: Mitigating Low Productivity on African Smallholder Farms [2]	AAAI	2022	ML	Clustered ensembles	Handles diverse climates; lower error vs. seq models	Cluster label quality sensitive	Forecasts stress/yield; not action policies
Optimal Greenhouse Sensor Placement via GBM Grid-Search [3]	Frontiers (Plant Science)	2022	ML	Gradient Boosting + feature eng.	Few sensors, low RMSE	Narrow variables (T/RH)	Two greenhouses; external validity unknown
Explainable Crop Recommendation with Ensemble ML [4]	Scientific Reports	2025	ML	Gradient Boosting + XAI (LIME)	High accuracy; interpretability	Static single-season advice	India-specific data; no dynamics
CropAndWeed Dataset: Multi-Modal Learning for Precision Weeding [5]	WACV	2023	DL	CNN detection/segmentation	Large aerial dataset; better weeding	Drone-only perspective	Ground/lighting generalization unclear

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Title	Venue	Year	Domain	Approach	Strength	Weakness	Limitations
iGrow: Autonomous Greenhouse Control with Deep RL [6]	AAAI	2022	DL/RL	Neural twin + deep RL	+10% yield; +92% profit	Calibration heavy; domain-specific	Greenhouse only; open-field generalization open
Deep Diagnosis: Real-time Apple Leaf Disease Detection [7]	CEA	2022	DL	Lightweight CNN (edge)	Real-time edge inference	Crop-specific	Cross-crop transfer not shown
Augmentation Invariance & Adaptive Sampling for Aerial Segmentation [8]	CVPR Workshops	2022	DL	Robust aerial segmentation	Higher mIoU; lighting/rotation robust	Training complexity	Limited to aerial; ground transfer uncertain
Vision–Language Few-Shot Tomato-Leaf Disease Detection [9]	AAAI (AI4SI)	2025	DL	VLM fine-tuning	Strong few-shot F1	Compute heavy	Tomato-only dataset
Nitrogen Management with RL and Crop Growth Models (CropGym) [10]	EDS (CUP)	2023	RL	PCSE/LINTUL-3 + RL	Open RL env; clear β trade-off	Single-season focus	Limited soil carry-over; narrow actions [b]
gym-DSSAT: a crop model turned into an RL environment [11]	Inria RR 9460 / arXiv	2022	RL	DSSAT + PDI + Gym	Daily control, high-fidelity sim	Heavy setup; maize-centric	Limited rotations; single-season emphasis

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Title	Venue	Year	Domain	Approach	Strength	Weakness	Limitations
Learning Long-Term Crop Management Strategies with CYCLES-GYM [12]	NeurIPS (D&B)	2022	RL	Cycles model + multi-year RL	Rotations, N-fixation, constraints	Higher config complexity	Fewer published studies than DSSAT/PCSE
Simulating Polyculture Farming to Learn Automation Policies [13]	IEEE TASE	2022	RL/Robotics	Polyculture simulation + RL	Diversity/irrigation policy	Setup complexity	Not crop-planning per se
DRL for Irrigation Scheduling with High-Dimensional Feedback [14]	PLOS Water	2023	RL	Sensor-driven DRL	Uses rich feedback	Sim/sensor dependence	Site-specific generalization open
RL vs. MPC on Greenhouse Climate Control [15]	CEA	2023	RL/Control	RL vs. MPC comparison	Benchmarks RL vs. MPC	Context-bound	Greenhouse focus
Constrained Multi-Objective RL Framework [16]	CoRL	2022	RL theory	Lagrangian/constrained RL	Principle for safety/sustainability	Algorithmic complexity	Application-specific tuning
State-Wise Safe RL: Survey [17]	IJCAI	2023	RL theory	Safety constraints (survey)	Safety taxonomy	Not agri-specific	Mapping to farms required

[a] Dhillon et al. (Frontiers 2023) hybridizes LUE with RF to raise predictive skill, but still requires local calibration and does not output decisions.
[b] CropGym documentation and EDS paper position it as a single-season nitrogen scheduling benchmark with configurable β for profit vs. sustainability.

2.7 Focused comparison of RL environments for our use-case

Table 2.2 provides a focused comparison of the most relevant RL environments for our use-case.

Table 2.2: RL Environments: Fit to Pakistan Rice–Wheat, Multi-Year, Cost-Aware Goals

Environment	Crop model & scope	Action space (examples)	Soil/rotation carryover	Notable constraint support
CropGym (EDS 2023)	PCSE/LINTUL-3; mainly winter wheat, single-season	Fertilizer timing/dose (5-day)	Limited explicit soil N carryover	β cost–yield trade-off (no strict caps)
Gym-DSSAT (Inria 2022)	DSSAT; daily maize season; WGEN weather	Daily fert./irrigation	DSSAT modules, but rotation demos limited	Custom reward penalties; setup heavy
CyclesGym (NeurIPS 2022)	Cycles; multi-year rotation (e.g., maize–soy)	Crop type, planting date, fert., irrigation	Explicit soil C/N, N-fixation across years	Leaching/emissions/economic costs as constraints

2.8 Key takeaways for CyclesGym-PK

- **Why not CropGym/Gym-DSSAT for Pakistan?** Both prove RL feasibility but are predominantly single-season or crop-specific; Pakistan needs monsoon-aware rotations and explicit soil carryover.
- **Why CyclesGym?** Direct support for multi-year rotations, nitrogen fixation, and environmental constraints aligns with sustainability goals and economic realism for rice–wheat systems.

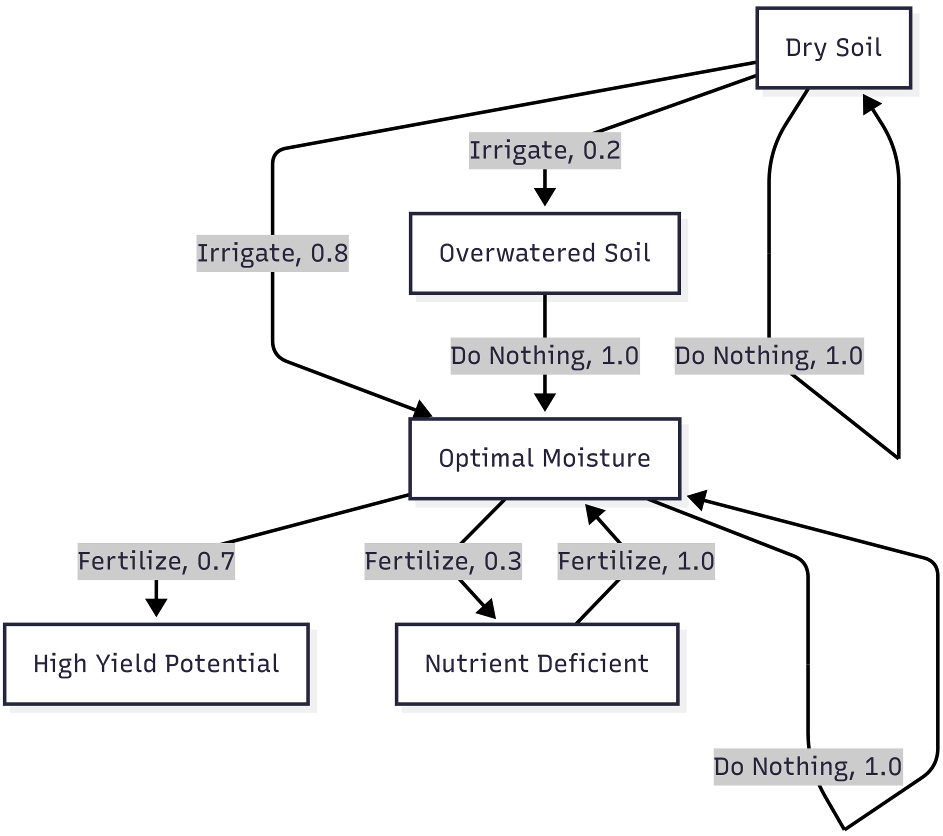

Figure 2.1: Decision flow for soil moisture and fertilization actions in the RL simulation.

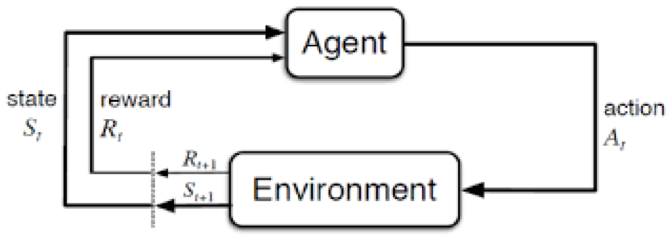

Figure 2.2: Reinforcement Learning agent–environment interaction loop.

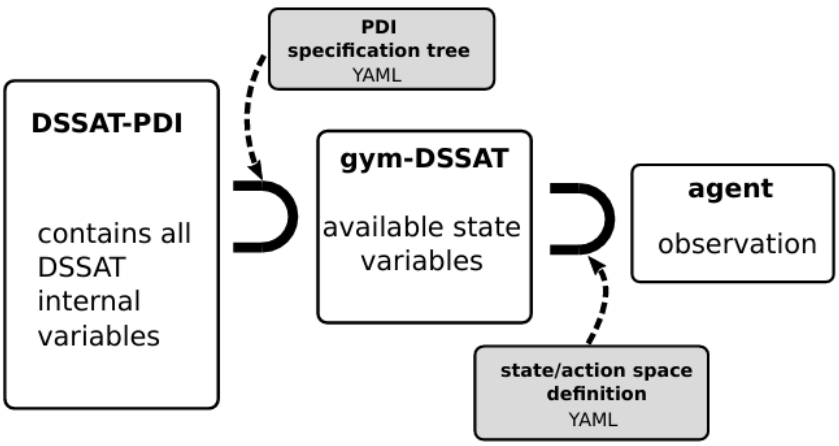

Figure 2.3: DSSAT–Gym integration: linking simulator internals to RL agents.

Chapter 3

Problem Statement and Objectives

3.1 Problem Statement

Existing RL environments target temperate, greenhouse, or single-season settings and do not transfer to Pakistan's monsoon climate, smallholder context, soils, and cost structures. There is no localized, multi-year, multi-objective RL framework for open-field Pakistan.

3.2 Research Questions

RQ1. How does CyclesGym compare with Gym-DSSAT and CropGym for simulating long-horizon, multi-crop, cost-aware policies?

RQ2. Which Cycles features (soil C/N dynamics, rotations) most enable sustainable profit under monsoon variability?

RQ3. How can learned policies be tailored to Pakistan's soils, weather, and economics for practical advisory use?

3.3 Research Objectives

O1. Build **CyclesGym-PK**: inject monsoon weather series, Indus-plain soils, and local cultivars.

O2. Design a cost-aware reward to balance PKR profit, water use, and N losses (leaching).

O3. Train PPO policies for weekly irrigation/fertilization; export readable rule tables.

O4. Evaluate robustness under dry/normal/wet years, soil PAW shifts, and price shocks.

Chapter 4

Research Methodology

4.1 Design Overview

We formulate weekly farm management as an MDP: state includes soil moisture/N, crop stage, weather summaries, and cumulative inputs; actions include irrigation (mm) and fertilizer (kg); reward equals revenue minus input costs and environmental penalties. Policies are learned via PPO with conservative updates (visualized in Figure 4.1).

Figure 4.1: Proximal Policy Optimization (PPO) update cycle.

4.2 Data Sources and Integration

- **Weather:** 30-year daily monsoon series (PMD-derived or reanalysis).
- **Soils:** Indus-plain textures and PAW from FAO-style profiles.

- **Crops:** IRRI-6 rice, Sehar-06 wheat phenology and N responses.
- **Economics:** PBS price bulletins (crop prices), market/fertilizer costs, irrigation energy tariffs.

These data sources are summarized in Table 4.1.

Table 4.1: Key Variables and Sources

Block	Variables (examples)	Source
Weather	daily rain, Tmin/Tmax, solar rad	PMD / reanalysis
Soil	texture, PAW, organic C, initial N	FAO-style profiles
Crops	cultivar params (gdd, LAI, N uptake)	IRRI/PARC literature
Economy	crop price (PKR/t), fertilizer and water cost	PBS / local bulletins

4.3 Environment Configuration (CyclesGym-PK)

- **Rotation:** Rice → Wheat, weekly decision step.
- **State:** soil moisture index, soil mineral N, crop stage, cumulative fert/irrigation, 1–2 week weather summaries.
- **Actions:** irrigation $\in \{0, 25, 50\}$ mm; fertilizer dose decisions at key growth stages.
- **Reward:**

$$R_t = \underbrace{\text{Revenue}_t - \text{WaterCost}_t - \text{FertCost}_t}_{\text{Profit}} - \beta_1 \cdot \text{Leaching}_t - \beta_2 \cdot \text{OverIrrig}_t$$

- **Algorithm:** PPO (clipped policy updates, $\gamma = 0.99$, entropy regularization).

4.4 Evaluation Protocol

- **Scenarios:** dry / median / wet monsoon years; soil PAW ±10%; price shocks (urea +15%, rice price -10%).
- **Baselines:** standard agronomic schedules for rice/wheat.
- **KPIs:** yield (t/ha), irrigation (mm), net profit (PKR/ha), N-use efficiency (%), leaching (kg/ha) (see Table 4.2 for details).

Table 4.2: KPIs and Acceptance Targets

KPI	**Target / Rationale**
Yield (t/ha)	Non-inferior to baseline in median year
Irrigation (mm)	Reduced vs. baseline without yield loss
Net profit (PKR/ha)	Increased or stable under price shocks
N-use efficiency (%)	Higher vs. baseline
Leaching (kg/ha)	Below threshold (e.g., 40 kg/ha/yr)

Chapter 5

Proposed Work

5.1 Work Packages (WPs)

WP1: Environment Localization — weather/soil ingestion; cultivar params; baseline yield validation.
WP2: Reward Engineering — profit & sustainability terms; β sensitivity grid; constraint tuning.
WP3: Policy Learning — PPO training; ablations (state features; action granularity).
WP4: Robustness and Reporting — stress tests; tables/plots; policy-to-rules export.

5.2 Model Architecture and Training

Two hidden layers (64 units, `tanh`); separate heads for discrete irrigation and staged fertilization; advantage normalization; parallel rollouts.

5.3 Deliverables

CyclesGym-PK code/configs; trained policies; baseline comparison tables; rule-table export for advisory app.

Chapter 6

Research Timeline

The proposed timeline is detailed in Table 6.1, with a corresponding Gantt chart shown in Figure 6.1.

Table 6.1: Major Tasks and Deliverables

#	WP	Task	Deliverable	Months
1	WP1	Localize weather/soil/cultivars; baseline yield check	CyclesGym-PK v1	1–3
2	WP2	Design reward; set β; constraint limits	Reward spec & tests	3–5
3	WP3	Train PPO; ablations; hyperparam search	Policy ckpts & logs	5–6
4	WP4	Robustness (weather/soil/price); KPIs	Results tables/figures	6–7
5	WP4	Report & rule-table export	Synopsis + appendix	4–7

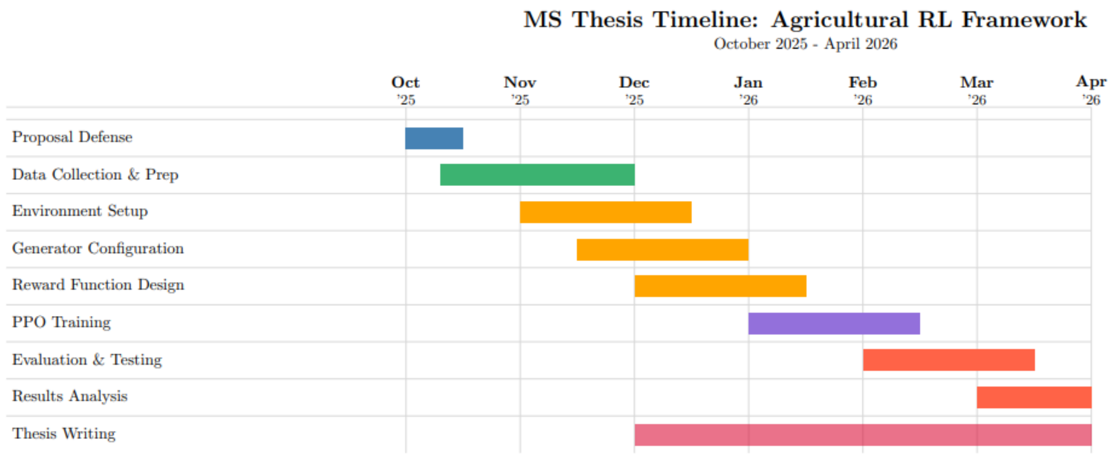

Figure 6.1: MS Thesis Timeline: Agricultural RL Framework (October 2025 – April 2026).

Chapter 7

Expected Contributions

- First Pakistan-focused, multi-year RL framework for open-field crop management.
- Demonstration of cost-aware, sustainability-constrained policy learning for rice–wheat rotations.
- Reproducible pipeline—data adapters, reward templates, policy export—for adaptation to other regions/crops.

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